

Aaditya Kalwani

Aerospace Engineering student focused on machine learning applied to physical engineering systems. Currently researching battery state-of-charge estimation toward publication. Comfortable across the stack: neural networks from scratch in NumPy, C# object-oriented simulations and computational tools for constrained engineering problems. Hands-on UAV 'design and make' experience (Team Bath Drones).

Experience & Technical Projects

Undergraduate Researcher -- Battery State-of-Charge Estimation, University of Bath | Jun '26 - Present

- **Developing a first-principles machine learning model** to estimate lithium-ion battery State of Charge by fusing ultrasound acoustic sensing with coulomb counting, targeting journal publication
- **Implemented a multilayer perceptron from scratch** in NumPy (forward pass, hand-derived backpropagation, training loop) before adopting frameworks, to ground the model in first principles
- Cut SOC estimation error to 0.64% by fusing displacement and ultrasound sensing to correct hysteresis, versus ~4% for coulomb counting alone: a **6x improvement** over the standard method.
- Contributing to the **Energetic (Horizon Europe)** project; paper targeted for peer review in Aug '26

Aerodynamics Division Lead -- Team Bath Drones, Design & Make Competition | Nov '25 – Present

- **Engineered an autonomous UAV** under strict objectives and material constraints, and **successfully delivered the Critical Design Review** with wing design choices and 400g weight reduction.
- Optimised aerodynamic efficiency: evaluated Clark-Y and NACA0015 profiles for a 30% increase in lift-to-drag ratio; co-led the 'tadpole' fuselage configuration and CAD model

Tech Strategy & Product Development Intern -- Fifth-9 (onsite, Canart Wharf) | Jul & Aug '25

- **Full Stack Development:** Architected and deployed 'PlayDay', a React/Node.js travel-planning engine, integrating the Google Gemini API for real-time, generative itineraries within a 5-day sprint
- **Strategic Analysis:** Formulated technical pitch decks and presented go-to-market strategies on emerging technologies (AI in Banking, Post-Quantum Cryptography) directly to company directors.

Founder -- STEM Tutoring Agency, + Private tutor, + ETutorYou | Sep '23 - Present

- Managed a tutoring portfolio for 15+ students across 3 tutors. Improved student performance by >3 grades via **AI-automation**, personalised feedback and targeted problem-set derivation.

Selected Engineering Projects

- Ultrasonic measurement system (Arduino + HC-SR04, MATLAB): calibrated a low-cost sensor to 1.0 mm RMSE (vs 4.3 mm spec) via filtering and regression (60% less noise); mapped its $\pm 18^\circ$ beam FOV and drove a state-machine monitor with a variance-based motion-rejection filter
- Ping-pong launcher (Y1 group design, 6 people): authored the winning launch-mechanism concept (slider-crank hammer), selected over five (weighted decision matrix). Team delivered full CAD, quick-return kinematics (27.8 m/s predicted) & 3D-printed build package
- Planetary gearbox (**Inventor**): built a 5:1 epicyclic assembly. Drove the carrier geometry from a projected base sketch; validated the 5:1 motion with joint constraints, materials, animated exploded view suitable for presenting to an outside stakeholder

Education

BEng Aerospace Engineering (Pred. First-Class Honours), University of Bath | Sep '25 - Present

- **Year 1 average: 74%:** Thermofluids (88%), Mechatronics (71%), Design (71%), Mathematics (70%)

Peter Symonds College | A-Levels: A*AAB in Maths, Physics, F. Maths, Comp Sci | Sep '23 - Jun '25

- 2x MEI National Maths Comp. | Silver, Senior Maths Challenge | Bronze, British Physics Olympiad

Gordon's School, Woking | GCSE: 6x Grade 9s inc. English & Further Maths, 5x 8s | Sep '21 - Jun '23

Selected Technical Skills

- **Languages:** English, Hindi | Python (NumPy, Pandas, ML (PyTorch, LSTM/MLPs)), C# (.NET/OOP), React
- **Engineering Tools:** CAD (Autodesk Inventor), MATLAB, Git+GitHub, Arduino / embedded